

JAVA LAB RECORD

Student Performance Analysis System

Experiment No.	04
Date	20/09/26
Title	Student Performance Analysis System Using Java

1. Objective / Aim

To develop a Java program using Object-Oriented Programming concepts to analyze a student's academic performance. The program accepts the student's roll number, name and marks of five subjects and calculates total marks, average marks, highest mark, lowest mark, percentage, grade, result and performance remark.

2. System Requirements

2.1 Hardware Requirements

- Computer or laptop
- Minimum 4 GB RAM
- Minimum 1 GHz processor
- At least 500 MB free storage
- Keyboard and monitor

2.2 Software Requirements

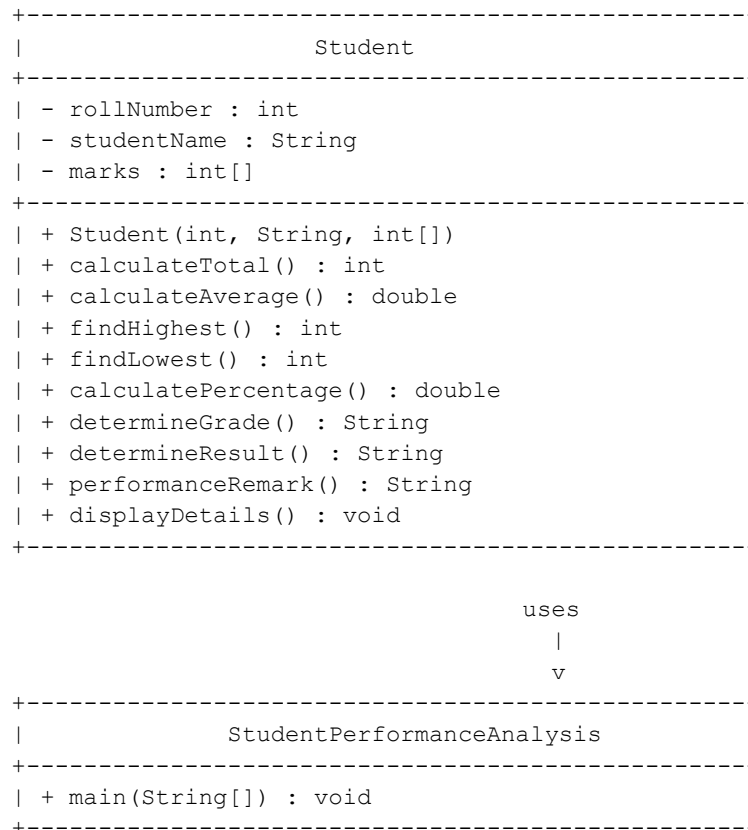
- Operating System: Windows / Linux / macOS
- Java Development Kit (JDK): JDK 17 or above
- Java compiler (javac)
- Java Runtime Environment (JRE)
- Any text editor or IDE such as Notepad, VS Code, Eclipse or IntelliJ IDEA

3. Algorithm / Logic Description

1. Start the program.
2. Create a Scanner object to accept input from the user.
3. Read the student's roll number and student name.
4. Create an integer array to store marks of five subjects.

5. Read and store the five subject marks in the array.
6. Create a Student object by passing roll number, name and marks to the constructor.
7. Calculate total marks by adding all marks in the array.
8. Calculate average marks by dividing total marks by the number of subjects.
9. Find the highest and lowest marks using the marks array.
10. Calculate percentage using the total marks and maximum possible marks.
11. Determine the grade using if-else conditions.
12. Determine PASS or FAIL based on the percentage.
13. Generate a performance remark based on the percentage.
14. Trim extra spaces from the student's name and convert it to uppercase.
15. Display the complete student performance report.
16. Close the Scanner and stop the program.

4. UML Class Diagram



Note: The Student class stores the student data and contains the methods used for performance calculations. The StudentPerformanceAnalysis class contains the main() method and creates the Student object.

5. Source Code

```
import java.util.Scanner;

class Student {

    int rollNumber;
    String studentName;
    int[] marks;

    // Constructor
    Student(int rollNumber, String studentName, int[] marks) {
        this.rollNumber = rollNumber;
        this.studentName = studentName;
        this.marks = marks;
    }

    // Calculate total marks
    int calculateTotal() {
        int total = 0;

        for (int mark : marks) {
            total += mark;
        }

        return total;
    }

    // Calculate average marks
    double calculateAverage() {
        return calculateTotal() / (double) marks.length;
    }

    // Find highest mark
    int findHighest() {
        int highest = marks[0];

        for (int mark : marks) {
            highest = Math.max(highest, mark);
        }

        return highest;
    }

    // Find lowest mark
    int findLowest() {
        int lowest = marks[0];

        for (int mark : marks) {
            lowest = Math.min(lowest, mark);
        }

        return lowest;
    }

    // Calculate percentage
```

```

double calculatePercentage() {
    int total = calculateTotal();
    return (total / (marks.length * 100.0)) * 100;
}

// Determine grade
String determineGrade() {
    double percentage = calculatePercentage();

    if (percentage >= 90)
        return "A+";
    else if (percentage >= 80)
        return "A";
    else if (percentage >= 70)
        return "B";
    else if (percentage >= 60)
        return "C";
    else if (percentage >= 50)
        return "D";
    else
        return "F";
}

// Determine result
String determineResult() {
    if (calculatePercentage() >= 50)
        return "PASS";
    else
        return "FAIL";
}

// Performance remark
String performanceRemark() {
    double percentage = calculatePercentage();

    if (percentage >= 90)
        return "Excellent Performance";
    else if (percentage >= 80)
        return "Very Good Performance";
    else if (percentage >= 70)
        return "Good Performance";
    else if (percentage >= 60)
        return "Average Performance";
    else if (percentage >= 50)
        return "Needs Improvement";
    else
        return "Poor Performance";
}

// Display complete student details
void displayDetails() {

    String formattedName = studentName.trim().toUpperCase();

    double percentage = calculatePercentage();

    // Round percentage to two decimal places
    double roundedPercentage = Math.round(percentage * 100.0) / 100.0;

    System.out.println("\n===== STUDENT PERFORMANCE REPORT =====");
    System.out.println("Roll Number      : " + rollNumber);
    System.out.println("Student Name      : " + formattedName);
}

```

```

        System.out.println("Name Length      : " + formattedName.length());

        System.out.println("Marks:");
        for (int i = 0; i < marks.length; i++) {
            System.out.println("Subject " + (i + 1) + "      : " + marks[i]);
        }

        System.out.println("Total Marks      : " + calculateTotal());
        System.out.println("Average Marks    : " + calculateAverage());
        System.out.println("Highest Marks   : " + findHighest());
        System.out.println("Lowest Marks    : " + findLowest());
        System.out.println("Percentage     : " + roundedPercentage + "%");
        System.out.println("Grade          : " + determineGrade());
        System.out.println("Result         : " + determineResult());
        System.out.println("Performance    : " + performanceRemark());

        System.out.println("=====");
    }
}

public class StudentPerformanceAnalysis {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Roll Number: ");
        int rollNumber = sc.nextInt();

        sc.nextLine(); // Consume newline

        System.out.print("Enter Student Name: ");
        String studentName = sc.nextLine();

        int[] marks = new int[5];

        System.out.println("Enter marks for 5 subjects:");

        for (int i = 0; i < marks.length; i++) {
            System.out.print("Subject " + (i + 1) + ": ");
            marks[i] = sc.nextInt();
        }

        // Create Student object
        Student student = new Student(rollNumber, studentName, marks);

        // Display performance report
        student.displayDetails();

        sc.close();
    }
}

```

6. Completion and Execution Commands

Step 1: Save the program with the file name:

StudentPerformanceAnalysis.java

Step 2: Open Command Prompt / Terminal in the folder where the Java file is saved.

Step 3: Compile the program using:

```
javac StudentPerformanceAnalysis.java
```

Step 4: Run the program using:

```
java StudentPerformanceAnalysis
```

Step 5: Enter the roll number, student name and marks for five subjects when prompted.

7. Output

Sample execution:

```
Enter Roll Number: 101
Enter Student Name: Rahul
Enter marks for 5 subjects:
Subject 1: 85
Subject 2: 92
Subject 3: 78
Subject 4: 88
Subject 5: 95

===== STUDENT PERFORMANCE REPORT =====
Roll Number      : 101
Student Name     : RAHUL
Name Length      : 5
Marks:
Subject 1       : 85
Subject 2       : 92
Subject 3       : 78
Subject 4       : 88
Subject 5       : 95
Total Marks     : 438
Average Marks   : 87.6
Highest Marks   : 95
Lowest Marks    : 78
Percentage      : 87.6%
Grade           : A
Result          : PASS
Performance     : Very Good Performance
=====
```

8. Completion

Thus, the Student Performance Analysis System was successfully implemented and executed in Java. The program demonstrates the use of classes, objects, constructors, arrays, methods, loops, conditional statements and basic String methods to analyze and display a student's academic performance.